

Integrating Retrieval-Augmented Generation and Thematic NLP for Vaccine Confidence Modeling in Alaska

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Abstract

Vaccine misinformation poses a significant public health threat, particularly in communities with varying levels of vaccine confidence. This study investigated vaccine hesitancy across Alaska's diverse communities by triangulating public sentiment from social media with individual beliefs gathered through qualitative interviews. The aim was to explore how online discourse influences vaccine-related decision-making and to develop tools for real-time misinformation detection.

We employed a mixed-methods approach, analyzing 1,300 Alaska-specific tweets and conducting 87 semi-structured interviews across urban and rural communities. A Retrieval-Augmented Generation (RAG) system was developed, integrating the context-rich LLaMA-2-7B model with the efficient T5-Base model to balance accuracy and computational performance. The system used sentence embeddings and FAISS-based similarity search to identify misinformation themes and generate context-aware responses grounded in real-world data.

Sentiment analysis revealed that rural social media posts exhibited significantly higher negativity and misinformation (55.6% negative sentiment) compared to urban posts. In contrast, interview data reflected more balanced and nuanced attitudes toward vaccination. Thematic analysis identified systemic distrust and personal beliefs, particularly among Indigenous and rural populations, as key drivers of hesitancy. Model evaluation showed that LLaMA-2-7B outperformed T5-Base in contextual accuracy, while T5-Base offered faster but occasionally less accurate responses.

By combining AI-driven insights with ethnographic data, this study highlights the divergence between online narratives and lived experiences. The proposed framework offers a scalable, real-time method for detecting misinformation and informing culturally responsive public health messaging. Future work will focus on optimizing system efficiency and collaborating with digital platforms to reduce the spread of viral misinformation.

Highlights

- Retrieval-Augmented Generation system detects vaccine misinformation using real-world social media data.
- Combined Llama-2 and T5 models for balanced accuracy and response speed.
- Sentiment analysis revealed distrust-driven negativity, especially in rural areas.
- Interview sentiment diverged from social media and showed more neutral attitudes.
- Triangulating NLP and ethnographic data improves public health insights.

Introduction

Vaccine hesitancy continues to pose a critical public health challenge worldwide^{1,2}, with rural and underserved populations like those in Alaska.³ Here, geographic isolation, historical trauma, particularly

among Indigenous populations, and distinct sociopolitical attitudes toward government and health autonomy shape divergent perceptions of vaccination. In addition, these communities often face intersecting barriers to vaccine uptake, including limited healthcare access, socio-cultural resistance, and deeply rooted institutional mistrust.⁴ This has been compounded by the COVID-19 pandemic⁵, which brought a proliferation of misinformation through social media platforms that outpaced public health responses.⁶ Social media algorithms are designed to maximize user interaction by amplifying emotionally charged and polarizing content, creating feedback loops that reward controversy over accuracy.⁷ In this environment, vaccine myths, particularly those related to safety, efficacy, and side effects, can rapidly go viral, entrenching skepticism and distrust.⁸

A wide range of factors contribute to vaccine hesitancy, including psychological concerns⁹, cultural and religious beliefs¹⁰, political ideologies¹¹, and longstanding mistrust of health institutions.¹² In the United States (U.S.), these drivers are magnified in historically marginalized communities, where public health campaigns must contend with legacies of systemic discrimination and medical exploitation.¹³ This multifaceted landscape calls for context-specific research that accounts for the structural, cultural, and informational dynamics at play.

Artificial intelligence (AI) offers powerful tools to investigate and address vaccine hesitancy at scale. Natural language processing (NLP) models, in particular, have been used to analyze health-related discourse, detect misinformation, and assess public sentiment across large datasets. While lexicon-based tools like VADER offer speed and simplicity, transformer-based models, such as DistilBERT and CardiffNLP's RoBERTa, enable more accurate sentiment classification in informal, short-form text like tweets.¹³⁻¹⁵ Retrieval-Augmented Generation (RAG) models represent a recent advancement in NLP, combining document retrieval with generative modeling to produce context-aware responses grounded in external evidence.^{15,16} These models have demonstrated effectiveness in applications such as fact-checking and misinformation detection, though their use in localized public health contexts remains limited.¹⁷

This study applies a novel, AI-augmented framework to investigate vaccine hesitancy in Alaska. We combine sentiment analysis of geotagged Twitter data with qualitative thematic analysis of in-depth interviews, offering both macro-level insights into public discourse and micro-level understanding of individual beliefs. Our approach incorporates a dual-model RAG system, using T5-Base for efficient, factual outputs and LLaMA-2-7B for nuanced, context-rich generation, alongside FAISS vector indexing and transformer-based sentiment classification. This study aims to understand vaccine hesitancy within Alaska's diverse communities by triangulating public sentiment (via social media) and individual beliefs (through qualitative interviews, combining technological insights with ethnographic data to create a multi-layered approach to vaccine misinformation detection. The analysis is framed within a novel Grounded Theory that integrates historical trauma, systemic distrust, and cultural beliefs as foundational drivers of vaccine hesitance. This theory was specifically suggested to address the unique sociopolitical and cultural dynamics present in Alaska, particularly among indigenous and rural populations. By

emphasizing the local context and community-driven narratives, the study aims to offer a more comprehensive understanding of vaccine hesitancy that builds upon relevant public health models such as the Theory of Change, offering targeted insights for culturally informed public health messaging.

Methods

Sampling and inclusion criteria

Participants were adult residents of Alaska (18+) with varying levels of vaccine hesitancy. Purposive sampling ensured representation across gender, race, geography (urban vs. rural), and hesitancy levels (hesitant, undecided, vaccinated). The sample included diverse ethnic groups, such as Native Alaskan, White, African American, Hispanic, and Asian residents. The novel grounded theory framework identifies systemic distrust as a central factor influencing vaccine hesitancy, particularly among indigenous communities, rooted in historical mistrust of government and healthcare institutions.¹⁸ A total of 87 semi-structured interviews were conducted via Zoom between July and November 2024, each lasting 30–40 minutes.

Data collection

Social media data were sourced from a publicly available Kaggle dataset¹⁹ and filtered for tweets geotagged within Alaska. Posts included were chosen based on keywords explicitly referencing vaccine-related topics, hesitancy, safety concerns, or COVID-19-related terms.

Data analysis

Interview transcripts were coded thematically to identify patterns in hesitancy and information sources across rural and urban populations.

Approximately 1,300 tweets were preprocessed for sentiment and misinformation analysis. Preprocessing included removing URLs, mentions, special characters, and emojis, expanding contractions, lower casing text, and lemmatization using the NLTK library. Tweets were geotagged to distinguish rural and urban regions. Sentiment analysis was performed using the CardiffNLP/twitter-roberta-base-sentiment model²⁰, fine-tuned for Twitter content. Tweets were tokenized using Hugging Face's AutoTokenizer²¹ and truncated to 512 tokens. Sentiments were classified into *positive*, *neutral*, or *negative* categories. For long-form interview transcripts, summaries capped at 500 tokens were generated using GPT-4-Turbo²² prior to classification with the same model. We used three complementary methods to detect misinformation in tweets: Keyword filtering flagged posts with known false claims; Vector-based retrieval compared tweet embeddings to misinformation examples using FAISS indexing^{23,24}; and semantic classification grouped flagged content into key themes like vaccine safety fears, political misinformation, and alternative medicine claims.²⁵ Together, these approaches enabled accurate detection across varied misinformation types in vaccine-related discussions. The first

RAG framework combined a T5-Base model¹⁶ with dense retrieval, and the second other integrated the LLaMA-2-7B model²⁶ for greater generative depth. Sentence embeddings of social media content were stored in an FAISS index for retrieval. The top 100 relevant posts were used to condition the generation phase. T5-Base generated concise output efficiently, while LLaMA-2-7B produced more nuanced responses for complex queries. The integration of these two data sources (qualitative social media data and qualitative interview data) enabled triangulation of findings, providing a more comprehensive understanding of the multifaceted drivers of vaccine hesitancy. This mixed-method approach, guided by a novel grounded theory, combines AI-driven analysis with ethnographic data to explore public sentiment and lived experiences, offering a richer perspective on the factors shaping vaccine decisions in Alaska. Fig. 2 illustrates the architecture of our RAG model.

Ethics

Ethical approval was obtained from the Institutional Review Board at Rutgers University (IRB #Pro2023002010, dated 04.12.2024). Informed consent was obtained from all participants. Interview data were anonymized. Social media data were collected from publicly available Twitter posts per the platform's terms of service, with no personally identifiable information stored.

Results

Sentiment analysis findings

Sentiment classification revealed clear differences in public discourse surrounding COVID-19 vaccination across Alaska’s rural and urban populations. Social media data consisted of approximately 1,300 geotagged tweets related to vaccines. Tweets were categorized into positive, neutral, or negative sentiment using the CardiffNLP/twitter-roberta-base-sentiment model.²⁰ Table 1 shows negative sentiment dominated in rural areas (55.56%), suggesting distrust-driven negativity, particularly in relation to vaccine mandates and governmental outreach. Rural tweets were more likely to express distrust, fear, or conspiracy-related concerns, often referencing vaccine passports or governmental mandates. In contrast, urban (48.23%) posts reflected a more polarized, politically charged narrative around the same themes. Neutral sentiment comprised 38.89% of rural and 41.79% of urban tweets, often referencing factual or news-related content. Positive sentiment appeared least frequently: 5.56% of rural posts and 8.96% of urban posts.

Table 1: Sentiment classification of vaccine-related tweets by urban/rural location

Sentiment	Urban/Rural	Percentage	Key Phrases
Positive	Rural	5.56%	"new", "https", "vaccine", "today", "absolute", "variant"
	Urban	8.96%	"https", "vaccine mandate", "vaccine", "booster", "pfizer", "great"
Neutral	Rural	38.89%	"passports", "https", "vaccine", "da", "im", "pfizer"
	Urban	41.79%	"https", "vaccine mandate", "vaccine", "mandate", "biden", "pfizer"
Negative	Rural	55.56%	"passports", "vaccine passports", "https", "vaccinated", "discriminatory", "pfizer"
	Urban	48.23%	"https", "vaccinated", "like", "moderna", "mandates", "vaccine mandates"

Interview sentiment, however, revealed more moderate views. As shown in Table 2, neutral sentiment dominated both urban (50%) and rural (48.65%) interviews. Negative sentiment was nearly equal across rural (43.24%) and urban (43.75%) respondents, and rural interviews actually showed slightly higher positivity (8.11% vs. 6.25%).

Table 2: Sentiment analysis of interview transcripts by urban/rural classification

Sentiment	Urban/Rural	Percentage
Negative	Urban	43.75%
	Rural	43.24%
Neutral	Urban	50.00%
	Rural	48.65%
Positive	Urban	6.25%
	Rural	8.11%

Thematic analysis findings

In line with the grounded theory approach, thematic insights emerged from the qualitative data (interviews and social media analysis), revealing complex, multi-layered factors influencing vaccine hesitancy in rural and indigenous communities. These insights were informed by both systemic distrust and personal/cultural beliefs, reflecting the intricate interplay of historical sociopolitical and cultural influences on health decision-making.

Systemic distrust and historical trauma

One of the core themes that emerged from both interviews and social media analysis was systemic distrust, particularly among indigenous populations. Thematic analysis highlighted how historical events, including colonial exploitation and medical mistreatment, contributed to a deep mistrust of the government and governmental healthcare systems.²⁷ This resonates with grounded theory's emphasis on contextual factors and historical understanding when constructing a theory.

Thematic patterns in the interviews also align with grounded theory's premise that data collection and analysis should highlight the structural forces affecting individual and group behaviors. The historical trauma influencing vaccine hesitancy points to a systemic distrust of public health systems, deeply embedded within the collective memory of indigenous communities. Grounded theories' emergent nature reveals this underlying theme without imposing pre-existing categories, showing that historical experiences shape modern-day vaccine decision-making.

Cultural identity and collective narratives

Another significant theme was the role of cultural identity in vaccine decision-making, which ties into personal and cultural beliefs. Rural participants, particularly those identifying as indigenous, viewed vaccine hesitancy not only as an individual decision but as a community-level issue. This collective perspective is crucial for understanding vaccine hesitancy within indigenous populations, where health decisions are framed in terms of community survival and sovereignty rather than individual autonomy (Table 3).

In grounded theory, the researcher's role is to remain open to emergent categories, such as collective versus individual decision-making. This insight is critical as it challenges the dominant individualistic framework often found in public health campaigns. The collective nature of health decisions in rural and indigenous communities reveals that public health messaging cannot simply focus on the individual's choice to vaccinate but must consider the community narrative of trust and cultural survival.

Misinformation and digital exposure

A key theme that emerged, particularly in rural areas, was the impact of misinformation on vaccine attitudes. Social media platforms like Facebook were identified as frequent sources of misinformation. This aligns with grounded theory's theoretical sampling approach as participants shared how misinformation spread through digital channels and influenced their vaccine beliefs.

Thematic analysis grounded in theory suggests that misinformation isn't just a product of technological platforms but is deeply entwined with cultural distrust of government and public healthcare institutions. This intertwined digital misinformation with systemic distrust creates a vicious cycle where misinformation feeds into already existing distrust, making it more challenging to influence vaccine uptake.

Political polarization and government distrust

The analysis also revealed that political polarization was a key influence on vaccine hesitancy, with differing views expressed in rural and urban settings. Rural participants tended to view government vaccine mandates as a form of political control, while urban participants tended to view government vaccine issues around personal freedoms. This polarization reinforces the grounded theory idea that political ideologies cannot be separated from public health behaviors.

Grounded theory emphasizes how social contexts like political climate shape the emerging patterns from the data. In this study, divergent views on government mandates reflect sociopolitical dynamics that directly impact vaccine hesitancy. The study’s theoretical framework uncovers how political narratives are internalized within rural communities, creating a unique form of systemic distrust tied to the perception of governmental interference.

Two primary dimensions that influence vaccine hesitancy emerged from qualitative data: institutional mistrust and political distrust. These themes reflected and expanded on the patterns observed in the sentiment data.

Perceived exploitation and historical trauma fueling hesitancy

Historical trauma, particularly among Indigenous rural communities, emerged as a prominent barrier to vaccine confidence. Participants referenced Western medical exploitation and colonial history. As a result, communities feel distrust of the federal government and worry that it is after profit, not their community’s well-being. Political polarization was a relevant issue for participants from rural and urban areas alike, although it is expressed in different ways. In urban areas, the discourse was framed around government mandates of vaccines infringing on personal freedom, while rural areas described this as being a way to control the population politically.

Misinformation pathways were shaped by digital exposure, and rural interviewees described frequent encounters with misinformation on platforms like Facebook, whereas urban participants demonstrated greater media literacy but still expressed concern over biased reporting.

Table 3: Themes related to trust and misinformation in vaccination

Main Theme	Subtheme	Description	Example Quote
Trust in the healthcare system	Historical mistrust in indigenous communities	Concerns rooted in historical mistreatment by authorities	" So, Alaska, we're spread out, difficult to access. And I'm curious if some of it is also a mistrust of government, especially when you're in remote, primarily Alaska Native villages. I wouldn't be surprised. There has been a lot of distrust in the federal government due to monetization. Western expansion. I think there's a lot of pain and trauma there. So, I think building trust in communities to bring in strangers to do these big vaccination clinics, I don't know if there's a trust."
Influence of social media	Spread of misinformation	False claims about the vaccine safety and side effects	"I think social media was a little too free with misinformation."
Political beliefs	Government distrust	Vaccines are seen as political control or an infringement of rights	"This I didn't trust, and I don't have trust in all the government."
Misinformation impact	Vaccine safety concerns/ Media influence	False beliefs about vaccines causing severe side effects	"What do you want to do? What do they want you to do? It depends on which platform you listen to, whether you have it, and whether you're foreign. But the mainstream is pushing people to be for it and to get all these vaccines. And now it's RSV, COVID, flu, whooping cough, and everything else."

Cultural values and safety fears as barriers to vaccination

Safety concerns were the most common personal barrier, cutting across rural and urban lines. Many expressed apprehensions about potential side effects, the perceived speed of vaccine development, and uncertainty about vaccine ingredients. One participant explained, "I think that really all the risks from the code vaccine came because of how much they rushed through the process to actually get it released." Others were skeptical of what goes into the vaccine. One interviewee said, "And not a lot of studies went into this vaccine, and I don't know all the ingredients in the vaccine, but this is just from what I hear, like online."

Natural immunity beliefs were more common among rural participants, with remarks like: "I feel like they could build a good natural immunity to it." This contrasted with urban participants who, while skeptical, were more likely to acknowledge vaccination benefits.

Cultural identity, especially among Indigenous respondents, shaped vaccine narratives as collective experiences of mistrust rather than individual decisions. One participant reflected, “For among native Alaskans, ... some who worried well, the government is experimenting on us.” This framing puts vaccination decisions within a broader cultural and historical context.

Table 4: Personal and cultural hesitancy themes from interviews

Main Theme	Subtheme	Description	Example Quote
Personal beliefs and values	Natural immunity preference	Preference for natural infection over-vaccination	“Do you believe that natural immunity acquired through infection is sufficient protection against COVID-19? Sometimes, yes.”
Social media influence	Viral misinformation	The rapid spread of false narratives through online platforms	“. . . during COVID, have you ever seen any kind of misinformation or some crazy information about vaccination, or could it be generally? Yeah, we would have. . . . there were all kinds of crazy things on Facebook, and people were pointing to those going.”
Vaccine concerns	Safety and side effects	Fears about adverse reactions and long-term effects	“So what do you think about vaccines generally . . . I think they’re unsafe.”
Cultural and social norms	Indigenous perspectives	Historical mistrust due to past medical exploitation	“We think that the government may again test a new vaccine. It is on us, so we cannot trust it. I wanted to ask you how you feel about it. UM, yeah, I feel like it’s sort of like. Like we’re Lab Rats being tested, you know.”

Table 5: Common Twitter discourse themes and frequencies

Theme	Frequency of mentions	Common hashtags/ Keywords	Example post
Misinformation	7.4%	#democratsdeliver, #togetherdeclaration, #covid19, https, COVID, people	EXCLUSIVE: Bucs receiver Antonio Brown obtained a fake COVID-19 vaccination card to avoid NFL protocols, according to hi...
Vaccine advocacy	21.5%	#democratsdeliver, #covid19, #omicron, https, covid, vaccinated	Guess I won the vax-lottery (AZ+Moderna). But now, which booster should I take? Another Moderna, so I\'m matched up (for ...
Fear-based content	9.2%	#vaccinemandate, #healthcareworkers, #astrazeneca, https, COVID, people	I know more people with serious adverse reactions to the vaccine than COVID-19....
Community support	7.0%	#covid19, #togetherdeclaration, #covid 19, https, COVID, help	This is strong. Thank you. Oklahoma National Guard & OK Governor. "Until a guardsman is activated under Title 10, they fo...

Platform-specific discourse patterns

Analysis of Twitter content revealed further distinctions in public sentiment and thematic framing, particularly around misinformation, advocacy, fear, and community narratives. 21.5% of tweets included vaccine advocacy content, often featuring hashtags like #democratsdeliver and references to vaccination success (Table 5). In contrast, 9.2% expressed fear-based content, typically focused on adverse reactions or perceived coercion.

Misinformation-themed tweets (7.4%) included politicized content, exaggerated claims, and links to discredited sources. Notably, community-support narratives comprised 7.0% of tweets, including posts thanking National Guard clinics and highlighting local efforts, aligning with rural interview themes of trust through community action.

These results indicate that social media narratives are often polarized or exaggerated compared to interview responses. While interviews explored distrust in nuanced ways, Twitter reduced concerns to binary hashtags and viral claims. This supports the study's use of platform-aware modeling for misinformation detection and community response strategies.

Theory of change

The multi-layered analysis reveals a theory of change where vaccine hesitancy is shaped by interconnected structural, sociopolitical, and individual-level factors. At its foundation, historical trauma and systemic distrust, particularly in Indigenous and rural communities, play a pivotal role in undermining vaccine confidence. These communities' collective memories of colonial exploitation and medical mistrust create deep-seated barriers to acceptance, which are further exacerbated by political polarization and misinformation. Social media amplifies these issues, fueling polarized narratives, misinformation, and fear-based content that shape public perception, especially in rural areas.

On the individual level, personal fears, cultural beliefs, and a preference for natural immunity influence decisions, particularly among rural participants who frame vaccination as a collective rather than an individual choice. Cultural identity and collective trauma contribute to the perception of vaccines as external threats rather than tools for community health.

To address vaccine hesitancy, it is essential to focus on long-term trust-building that acknowledges the historical, cultural, and sociopolitical realities of these communities. This requires not only platform-specific misinformation countermeasures but also culturally relevant messaging that resonates with both historical experiences and personal beliefs. An integrated, context-specific approach that combines AI-driven misinformation detection with ethnographic insights provides a pathway to improve vaccine uptake through targeted interventions that engage the structural, sociopolitical, and individual layers of influence.

The thematic insights derived from this study, grounded in the novel theory of systemic distrust and historical trauma, underscore the necessity of moving beyond the traditional public health framework. Public health messaging must be culturally sensitive and historically informed, accounting for the collective nature of health decisions in rural and indigenous communities, as well as the impact of misinformation spread through digital platforms. The grounded theory approach provides a robust methodology for uncovering the underlying drivers of vaccine hesitancy and informs the development of targeted, contextually appropriate interventions.

Discussion

This study revealed a significant divergence between public discourse on social media and individual perspectives shared in interviews. Social media content, particularly from rural areas, tended to amplify polarized and negative sentiment, while interviews reflected more balanced, reflective, and contextually grounded views. By combining computational tools with qualitative inquiry, this mixed-methods approach provided a nuanced understanding of vaccine hesitancy across urban and rural Alaska. The integration of RAG models, sentiment analysis, and thematic coding enabled the detection of misinformation trends as well as culturally rooted vaccine narratives. These findings highlight critical gaps between online narratives and lived experiences, pointing to the complex interplay of sociocultural, informational, and geographic factors influencing vaccine decision-making.

A key finding is the stark contrast between social media and interview sentiment. Twitter showed more polarization, especially in rural areas, while interviews revealed more moderate views, with neutral sentiment prevailing and rural participants slightly more likely to express positive perspectives.

This disparity suggests that social media platforms, particularly Twitter, serve as amplifiers of extreme sentiment rather than accurate mirrors of community-level beliefs. While interviews revealed concerns about vaccine safety, side effects, and mistrust, these were often accompanied by thoughtful reflections, conditional acceptance, or pragmatic considerations.²⁸ Social media, on the other hand, tended to collapse these nuances into viral, often alarmist narratives. This underscores the importance of supplementing social media surveillance with qualitative data to avoid overestimating public hostility and mischaracterizing the complexity of vaccine decision-making.²⁹

Across both data sources, distrust in government and healthcare systems emerged as a dominant theme, particularly among rural and Indigenous populations. Thematic analysis highlighted that many rural interviewees, especially from Alaska Native communities, framed hesitancy within a broader historical context of colonization and systemic marginalization. References to being “lab rats” or skepticism toward outside health authorities underscore how the collective memory of exploitation and disempowerment continues to shape vaccine perceptions today.

This aligns with previous research demonstrating that historical trauma is a persistent determinant of health behavior in Indigenous communities.^{30,31} However, the present study adds granularity by showing how these legacies interact with real-time political and media environments. While urban participants were also distrustful, their concerns more frequently revolved around political polarization and media bias rather than historical injustice.³² Thus, tailored public health messaging must not only address current misinformation but also contend with deeper-rooted narratives of marginalization and betrayal, particularly in rural and Indigenous settings.

The influence of social media, particularly Facebook and Twitter, emerged as a significant driver of vaccine attitudes.^{33,34} Rural interviewees frequently describe encountering and sometimes believing misinformation online. These narratives overlapped with tweets that referenced fake vaccine cards, exaggerated side effects, or political conspiracy theories.

Twitter content analysis reinforced this divide, showing that misinformation-themed tweets, while only 7.4% of the sample, were often highly politicized and emotionally charged. The prevalence of hashtags like #togetherdeclaration and the conflation of unrelated health threats (e.g., RSV, flu, COVID-19) point to an ecosystem where fear and confusion are actively circulated. By contrast, the more reflective tone in interviews suggests that personal conversations may serve as a corrective to online distortion.

Beyond institutional distrust, individual-level concerns, particularly around vaccine safety and natural immunity, were common. Both rural and urban interviewees expressed fears of side effects and distrust

in the rapid development of COVID-19 vaccines. However, belief in natural immunity was notably more common among rural participants.

Cultural identity also played a key role. Among Indigenous participants, vaccine hesitancy was framed not just as an individual choice but as a community-level issue tied to sovereignty, history, and cultural survival. This suggests that strategies to increase vaccine uptake must avoid individualistic framings and instead engage collective narratives, especially in Indigenous contexts. Public health campaigns that ignore these social dynamics risk reinforcing existing mistrust.

These findings carry several important implications for public health planning and vaccine promotion. First, they highlight the risks of over-relying on social media analytics to gauge public opinion. While valuable for detecting trends, social media data often reflect the loudest or most extreme voices, especially in politicized contexts like vaccination [40,41].

Second, the persistence of historical and systemic distrust, particularly among rural and Indigenous communities, demands long-term engagement strategies rooted in cultural humility, respect, and relationship-building. Public health institutions must invest in sustained partnerships that foster local leadership and community ownership of health initiatives.

Finally, the study underscores the need for platform-specific misinformation countermeasures. Community-driven media literacy initiatives, coupled with local storytelling and trusted messengers, may offer more effective pathways for addressing misinformation than top-down fact-checking alone.

The RAG pipeline proved effective for misinformation detection and tailored response generation. LLaMA-2-7B consistently produced more nuanced, contextually rich outputs than T5-Base, especially for complex queries involving cultural or historical themes. However, it required significantly more computational resources. T5-Base, while faster, occasionally repeated misinformation when exposed to frequently retrieved themes, an issue partially mitigated by post-processing and evidence filtering.

These findings align with recent work advocating hybrid NLP systems for misinformation detection.¹⁷ Our integration of dense retrieval via FAISS²³, sentiment classification using transformer models.²⁰ Culturally contextualized interviews are also a key component in our model.

Beyond computational performance, this study emphasizes the importance of data triangulation. Sole reliance on social media analysis risks overlooking the nuance and lived experiences that interviews provide. By integrating social and cultural narratives with large-scale textual analysis, our approach addresses this gap.

By employing grounded theory, this study allows thematic insights to emerge naturally from the data, ensuring that the analysis is context-specific and sensitive to the realities of Alaskan communities. The theory of systemic distrust is not imposed but is constructed based on the patterns revealed through both interviews and social media discourse. The emergent themes (historical trauma, cultural identity,

and misinformation) align with grounded theory's approach to data-driven theory development, offering a comprehensive understanding of vaccine hesitancy that incorporates both local contexts and digital influences.

The study has several limitations. The LLaMA-2-7B model, while effective, is resource-intensive and may not support real-time applications in constrained environments. In contrast, T5-Base, though faster, sometimes lacked contextual sensitivity or echoed common misinformation phrases. Additionally, synthesizing multiple tweets occasionally resulted in conflicting inputs, requiring additional filtering to ensure coherence. Sentiment models, though strong on short-form text, may overlook sarcasm or ambiguity in nuanced narratives.^{15,35}

Despite these limitations, the study provides a scalable framework for vaccine misinformation tracking and response. Real-time RAG systems, integrated into public health messaging workflows, could help agencies detect emerging misinformation trends and issue timely, evidence-based responses. Tailoring these systems with culturally informed data, as demonstrated here, further enhances their relevance and equity.

Conclusion

Vaccine hesitancy in Alaska reflects a dynamic interplay between historical experience, personal belief, and digital information flows. Rural and urban communities differ not only in the content of their concerns but also in how those concerns are expressed and shaped by social context. The thematic insights derived from this study, grounded in the novel theory of systemic distrust and historical trauma, underscore the necessity of moving beyond traditional public health frameworks. Public health messaging must be culturally sensitive and historically informed, accounting for the collective nature of health decisions in rural and indigenous communities, as well as the impact of misinformation spread through digital platforms. The grounded theory approach provides a robust methodology for uncovering the underlying drivers of vaccine hesitancy and informs the development of targeted contextually appropriate interventions. Going forward, efforts to improve vaccine confidence must move beyond combating misinformation and address the deeper structural and cultural roots of distrust.

Declarations

Ethical approval was obtained from the Institutional Review Board at Rutgers University (IRB #Pro2023002010, dated 04.12.2024). Informed consent was obtained from all participants. Interview data were anonymized. Social media data were collected from publicly available Twitter posts per the platform's terms of service, with no personally identifiable information stored.

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Figures

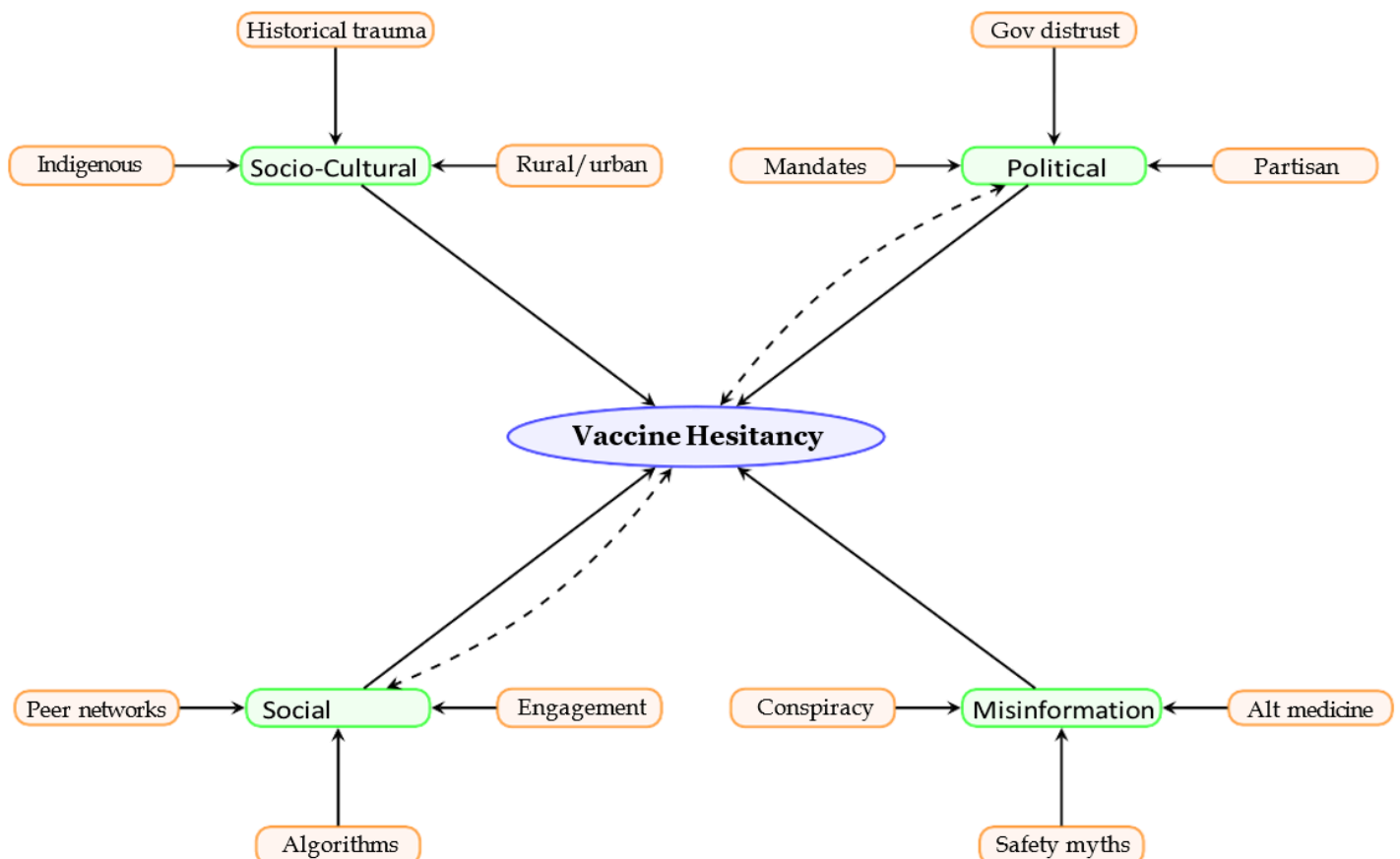


Figure 1

Factors contributing to vaccine hesitancy, categorized into socio-cultural, political, social media, and misinformation influences.

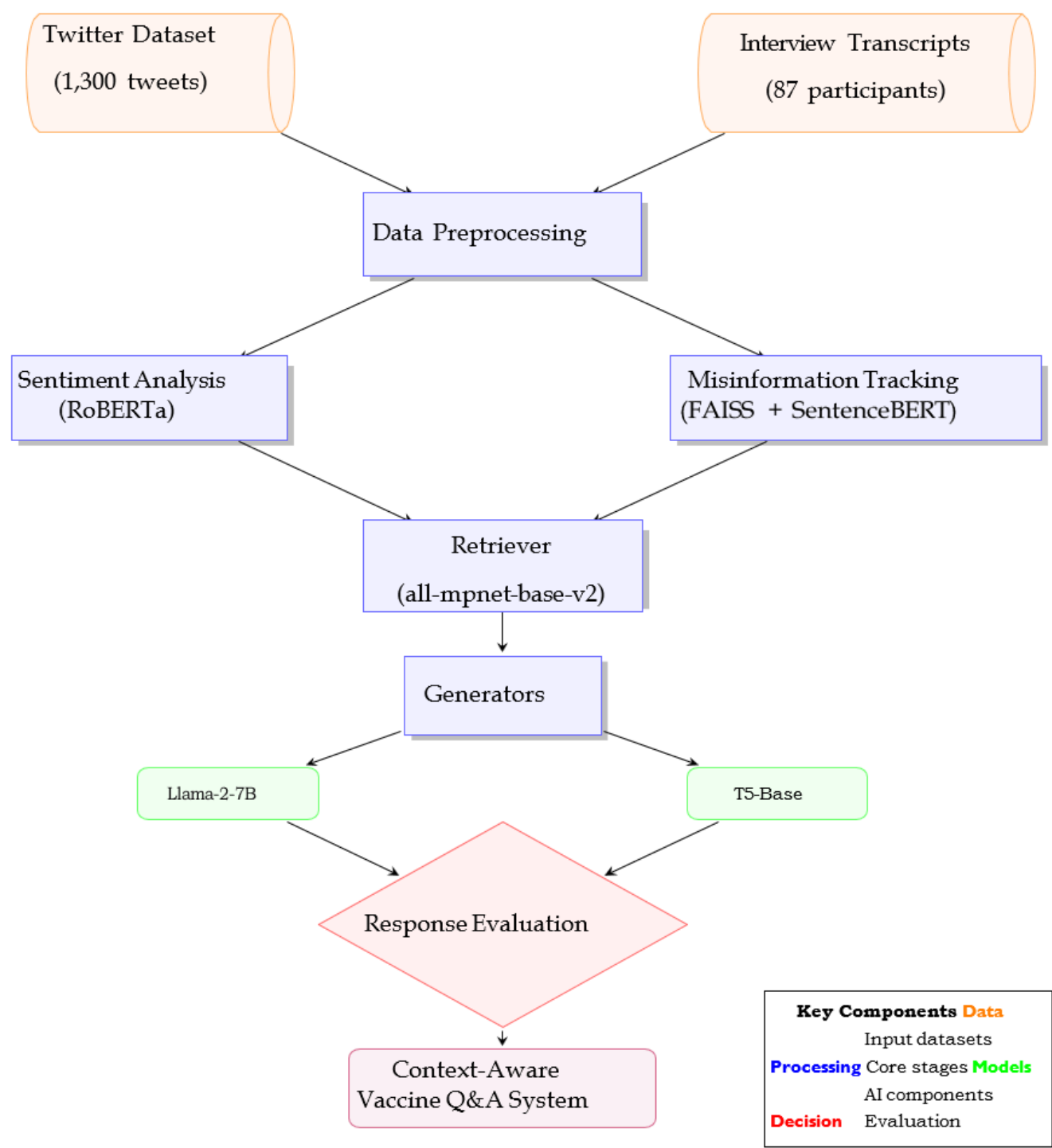


Figure 2

Compact architecture of the RAG system for vaccine misinformation analysis with optimized spacing.